

Total No. of Questions : 6]

SEAT No. :

P891

[Total No. of Pages : 2

BE/Insem.-18
B.E. (Mechanical)
REFRIGERATION & AIR CONDITIONING
(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Assume suitable data if necessary.*

Q1) a) Explain with neat diagram the working of Bell Coleman Cycle and Explain COP formula with TS diagram. [6]

b) Explain the desirable properties of refrigerant. [4]

OR

Q2) a) Write a note on hospital air conditioning. [6]

b) List out any four names of inorganic refrigerants with refrigerant number. [4]

Q3) a) Explain thermodynamic requirements of refrigerant-absorbent mixture used in vapour absorption refrigeration system. [4]

b) A vapour compression machine uses R12 refrigerant operates between temperatures limit of 36°C and -7°C. The compression is isentropic and there is no undercooling. Vapour is dry saturated at the entry of compressor. Find the COP of the system. Properties of refrigerant are as follows : [6]

Take $C_{pv} = 0.64 \text{ kJ/kgK}$

Sat. Temp	h_f	h_g	S_g
°C	kJ/kg	kJ/kg	kJ/kg.K
36	70.55	201.58	0.6836
-7	29.62	184.5	0.7001

P.T.O.

OR

Q4) a) Explain with diagram Lithium bromide vapour absorption refrigeration system. [6]

b) Draw T-s and P-h diagram for the effect of evaporator pressure on COP of simple vapour compression cycle. [4]

Q5) a) Explain two stage vapour compression with perfect liquid intercooler with schematic and p-h diagram. [6]

b) Define multi pressure system and explain necessity of multi pressure system. [4]

OR

Q6) a) What are the advantage of compound compression with intercooler over single stage compression. [4]

b) Explain the cascade refrigeration system with schematic and p-h diagram. [6]

